

Topic:-

Visit an ecosystem nearby. List the biotic and abiotic factors in it and show with pictures or sketches how they are dependent on each other.

Introduction:-

Nature is a magnificent, harmonious web where all living and non-living components interact continuously. An **Ecosystem** is a self-sustaining biological community consisting of living organisms interacting with their non-living physical environment. The term 'ecosystem' was first coined by English ecologist A.G. Tansley in 1935.

Every ecosystem on planet Earth is composed of two inseparable, interdependent parts: **Biotic factors** (all living beings like green plants, animals, insects, fungi, and bacteria) and **Abiotic factors** (all non-living physical elements like sunlight, water, air, soil, minerals, and temperature). Neither can exist without the other. Green plants cannot synthesize food without sunlight and water, while animals cannot breathe without oxygen produced by plants. For our Class 8 science project on Ecosystems, our student group visited a natural freshwater garden pond ecosystem near our school to observe, catalog, and sketch their interdependence.

❖ Two Main Components of an Ecosystem:-

- **Biotic (Living):** Producers (autotrophs), Consumers (heterotrophs), Decomposers (saprophytes).
- **Abiotic (Non-Living):** Sunlight, water, atmospheric gases (O_2 , CO_2), soil minerals, temperature.

Aim & Field Visit Setup:-

Aim: To visit a local garden pond ecosystem, identify and catalog all biotic and abiotic factors, and illustrate their mutual ecological interdependency through sketches and food web diagrams.

❖ *Objectives of the Field Study:-*

- To record abiotic physical parameters (water pH, temperature, clarity, soil type).
- To identify living producers, primary consumers, secondary consumers, and decomposers.
- To sketch an ecological food chain demonstrating solar energy flow through the pond community.

❖ *Study Site & Field Tools Carried:-*

- **Site Visited:** School Botanical Garden Freshwater Pond and wetland embankment.
- **Field Kit:** Field notebook, pen, magnifying hand lens, pH indicator paper, digital camera.



Field Visit Methodology & Observations:-

Our student team spent 2 hours quietly observing the pond ecosystem and recorded:

- **1. Abiotic Physical Factors Surveyed:**

- *Sunlight*: Bright unfiltered morning sunlight penetrating clear water up to 1.5 meters depth.
- *Water*: Freshwater pond measuring 25°C temperature; pH strip showed neutral 7.2.
- *Air*: Continuous atmospheric oxygen dissolving into water ripples for aquatic respiration.
- *Bottom Mud*: Rich dark organic silt providing anchor roots and mineral nutrients.

- **2. Biotic Living Factors Documented:**

- *Producers (Plants)*: Green filamentous algae (Spirogyra), floating duckweed, and water lilies.
- *Primary Consumers (Herbivores)*: Water snails, mosquito larvae, tadpoles feeding on green algae.
- *Secondary Consumers (Carnivores)*: Diving water beetles, minnow fish, green frogs, dragonflies.
- *Tertiary Consumer (Apex)*: Kingfisher bird diving from a branch to catch small fish.
- *Decomposers*: Soil bacteria and aquatic fungi feeding on dead sunken leaves at pond bottom.

Ecosystem Factors & Interdependence Table:-

The table below documents how biotic and abiotic factors interact and depend on each other:

<i>Factor Category</i>	<i>Specific Example</i>	<i>Dependency on Other Factors</i>	<i>Contribution to Pond Ecosystem</i>
Abiotic (Light)	Solar Sunlight	Reaches from Sun (Abiotic)	Powers photosynthesis in green algae and plants
Biotic (Producer)	Green Algae & Lilies	Needs sunlight, water, CO ₂ , mud minerals	Produces food glucose and releases dissolved oxygen
Biotic (Herbivore)	Tadpoles & Water Snails	Feeds on living green algae; needs oxygen	Controls algal blooms; provides food for fish
Biotic (Carnivore)	Green Frogs & Small Fish	Eats water insects and tadpoles; needs water	Controls mosquito insect population
Biotic (Decomposer)	Mud Bacteria & Fungi	Feeds on dead fallen leaves and fish waste	Breaks organic waste into mineral nitrates for plants

Scientific Analysis & Ecological Cycling:-

Our field observations clearly demonstrated the two great laws of ecosystem science:

- **1. Unidirectional Energy Flow:** Solar energy captured by green producers flows to herbivores (tadpoles) and then to carnivores (frogs and kingfishers). At each trophic step, nearly 90% of energy is lost as metabolic heat, and only 10% passes to the next level (Lindeman's 10% Law). Energy never flows backward!
- **2. Cyclic Movement of Matter (Nutrient Cycle):** Unlike energy, mineral nutrients never disappear. Carbon, nitrogen, phosphorus, and water cycle endlessly between abiotic air/soil and biotic living bodies. When plants and fish die, decomposers return their mineral atoms back to the bottom mud, where roots reabsorb them.

❖ Delicate Ecological Balance:-

If abiotic water is polluted with detergents or plastic, sunlight is blocked, algae die, oxygen vanishes, fish suffocate, and the entire pond collapses into a foul swamp!

Conservation & Environmental Protection:-

Pond ecosystems provide vital ecological services to human settlements:

- **Natural Rainwater Recharge:** Ponds soak up monsoon runoff, preventing localized flash floods and recharging underground drinking water aquifers.
- **Biological Mosquito Control:** A healthy pond teeming with frogs, dragonflies, and minnow fish consumes thousands of mosquito larvae daily, keeping neighborhoods disease-free.
- **Urban Biodiversity Oasis:** Ponds provide essential drinking water and nesting trees for migratory birds, butterflies, and bees in concrete cities.

❖ Action Steps Taken by Our School Science Club:-

- Organized a pond clean-up drive: removed 15 kg of discarded plastic bottles and snack wrappers.
- Planted native flowering shrubs along the pond bank to prevent mud erosion during rains.
- Erected a sign board: *'Do Not Throw Plastic or Idol Waste into Pond Water'*.

What We Learned & Conclusion:-

- An ecosystem is a self-sustaining web where biotic living beings and abiotic physical factors interact continuously.
- Green plants are the vital biological link converting abiotic solar energy and minerals into edible food.
- Decomposers act as nature's unsung recyclers, returning dead organic matter back into abiotic soil minerals.
- Protecting local wetlands and ponds from plastic waste and chemical pollution is essential to sustain life on Earth.

Conclusion:-

Our field visit to the garden pond ecosystem brought textbook ecology to vibrant life. Observing frogs, algae, water snails, and dragonflies interacting with sunlight, water, and mud proved that every living creature is connected to its abiotic surroundings. This study instilled in our class a deep reverence for the delicate balance of nature and inspired us to become active guardians of our local water bodies.

References:-

- Maharashtra State Board Class 8 General Science Textbook (Chapter 18: Ecosystems).
- Field Study Guide for Ecology & Freshwater Wetlands, Balbharti Pune.
- Photographs and Field Observations Log, Class 8 Nature Club.